🛡️ PhishGuard — Final Scope (MVP v1)

PhishGuard is a **privacy-first Chrome extension** that protects users from phishing and social engineering attacks in real time within web-based messaging platforms. It analyzes messages and URLs using **on-device NLP and heuristic risk scoring**, flags threats with explainable alerts, and ensures **zero message storage or transmission**.

🎯 1. Platform Coverage

Supported platforms:

- Gmail
- WhatsApp Web

(Additional platforms planned for future phases)

⚡ 2. Real-Time Message Scanning

Mechanism:

- Uses **DOM MutationObserver** to monitor incoming messages in real time
- Scans **only newly added message nodes**
- Prevents duplicate processing using **WeakSet tracking**

Detection Targets:

- **Urgency / pressure language**
  Example: “Act now”, “Account will be suspended”, “OTP expires in 2 minutes”
- **Sensitive information requests**
  Example: OTP, passwords, Aadhaar, credit/debit card details, personal data
- **Impersonation attempts**
  Example: bank, HR, government, support representatives
- **Linguistic anomalies**
  Unusual grammar, overly formal tone, phishing-like phrasing patterns
- **Embedded URLs**
  All links are automatically extracted and analyzed

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3. URL Analysis Engine

Every detected URL is analyzed for:

- **Homoglyph & typosquatting detection**
  (e.g., paypa1.com, g00gle.com)
- **Punycode / IDN decoding**
  Detects visually deceptive internationalized domains

- **Redirect chain resolution**
  Expands shortened URLs (bit.ly, tinyurl) to final destination

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4. Detection Engine (Core System)

A. Feature Extraction Layer

Each message is analyzed for:

- Urgency indicators
- Sensitive data requests
- Impersonation keywords (bank, HR, government, support)
- Linguistic anomalies
- URL presence and structure

B. On-Device ML Model

- Fine-tuned **XGBoost Classifier + TF-IDF + Feature engineering**
- Runs fully in-browser
- Performs phishing classification
- **No message leaves the device**

C. Risk Scoring Engine (Core Logic)

Each message is assigned a **risk score (0–100+)** based on combined signals.

Example Scoring Model:

+30 → urgency detected
+40 → sensitive info request (OTP/password)
+25 → suspicious / shortened URL
+20 → impersonation keywords
+15 → linguistic anomaly

Score Interpretation:

- **0–30** → Safe
- **30–60** → Suspicious
- **60–80** → High Risk
- **80+** → Critical

Final Score Composition:

- ML prediction confidence
- URL analysis results

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5. Alerts & Reporting

Risk-Scored Alerts

Each flagged message generates a popup containing:

- Risk level (Low / Medium / High / Critical)
- Risk score
- Clear explanation

Threat Transparency

Users see exactly why a message was flagged.

Example:

“This message requests your OTP and contains a suspicious shortened link”

User Actions

- ✅ Mark as Safe (false positive feedback)
- 🚨 Report phishing to:
    - PhishTank
    - Google Safe Browsing

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6. Performance Optimization

To ensure efficient browser performance:

- Scan **only newly added messages**
- Track processed nodes using **WeakSet**
- Apply **debouncing (300–500 ms)**
- Limit message length (e.g., max 500 characters)

Skip Processing:

- Media content
- Large static message threads
- Already processed nodes

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7. Privacy & Ethics

- ✅ All processing is **on-device only**
- ❌ No message storage
- ❌ No message logging

- ❌ No transmission of message content

9. Feedback & Learning

- Users can:
  - Mark messages as safe
  - Confirm phishing
- Feedback is used for:
  - Local model improvement
  - Future model refinement
- No raw messages are stored or transmitted

Architecture system

DOM Observer

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Message Extraction

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Feature Extraction

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ML Model (DistilBERT)

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URL Analysis Engine

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Risk Scoring Engine

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Alert System


Browser
WhatsApp Web
New message appears
content.js
MutationObserver
Extract text + URLs
Gmail
Email opened
PREDICT msg
Chrome Extension (Service Worker)
background.js
API proxy + log storage
POST /predict
Local FastAPI Server (127.0.0.1:8000)
app.py
TF-IDF + XGBoost
Returns probability
Model files
.pkl .txt
threshold, tfidf, keywords
prediction + probability
Hybrid scoring (content.js)
ML score
probability × 10
Rule boost
URL + keyword flags
Hybrid score
ML + (rule × 0.3)
score ≥ 5 → flag
Highlight
Red banner on msg
Toast alert
Top-right popup
Popup log
Stored in chrome.storage